13 Prove that the recurring decimal $0.4\dot{3}$ has the value $\frac{13}{30}$

Let x = 0.43333...

1 mark
Align recurring tails

10x = 4.33333...

Subtract: 10x – x = 4.33333... – 0.43333... = 3.9

9x = 3.9, so 90x = 39 and x = 39/90 = 13/30.

1 mark
13/30 proved

(Total for Question 13 is 2 marks)